

Meng'en Qin

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Summary

Hi, I'm Meng'en Qin, a statistics graduate from Henan University. My research journey began at Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms, working on computer vision and its interdisciplinary applications. Now, I'm focusing on the interpretability and hallucinations in MLLMs, fortunately working with Prof. Ruize Han and Dist. Prof. Song Wang from Shenzhen University of Advanced Technology.

Education

School of Mathematics and Statistics, Henan University

Kaifeng, China

B.S. IN STATISTICS

Sep 2021 - Jun 2025

- Faculty Ranking: top 8% and GPA: 87/100

Research Experience

Shenzhen University of Advanced Technology

Shenzhen, China

RESEARCH ASSISTANT

Sep 2025 - Sep 2026

- interpretability and hallucination in MLLMs (working with Prof. Youlu Xing, Prof. Ruize Han and Dist. Prof. Song Wang)

Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms

Henan, China

RESEARCH INTERN

Nov 2023 - Jan 2025

- computer vision and its interdisciplinary applications (work with Prof. Xiaohui Yang)

Honors & Awards

Nov 2023 **Provincial First Prize**, China Undergraduate Mathematical Contest in Modeling

Henan, China

Aug 2023 **National Second Prize (Top 0.5%)**, National College Student Statistical Modeling Competition

China

Aug 2023 **National Third Prize**, National College Students Social Practice and Science Contest on Energy Saving & Emission Reduction

China

Papers

MLLMs Hallucinate when Information Distribution Drifts in Synergy Heads

MENG'EN QIN, JUNYE CHEN, JUCHENG LIU, YINCHEN LIU, YOU LU XING, SONG WANG, RUIZE HAN[†]

2026

Training-Adaptive Convolutional Sparse Coding via Information Bottleneck for Robust Visual Representation

MENG'EN QIN, YINCHEN LIU, MINGXUAN CUI, YOU LU XING[†]

2026

Adaptive Compressed Residual: Redundancy-Aware Mechanism for Compact Multimedia Representation Learning

revision resubmitted to TMM

MENG'EN QIN*, YU SONG*, JING FANG, DEYU MENG, XIAOHUI YANG[†], LILI YANG[†]

2026

A³-FPN: Asymptotic Content-Aware Pyramid Attention Network for Dense Visual Prediction

Pattern Recognition

MENG'EN QIN, YU SONG, QUANLING ZHAO, XIAODONG YANG, YINGTAO CHE, XIAOHUI YANG[†]

2026

StomaD²: An All-in-One System for Intelligent Stomatal Phenotype Analysis via Diffusion-Based Restoration Detection Network

QUANLING ZHAO*, MENG'EN QIN*, YANFENG SUN, YUAN MIAO[†], XIAOHUI YANG[†]

2025

Significance and Stability Analysis of Genotype-Environment Interaction using GxEStat

MENG'EN QIN, ZHE LI, HUI HUANG, XIHONG LIU[†]

2026

OS-MSWGBM: Intelligent Analysis of Organic Synthesis Based on Multiscale Subtraction Weighted Network and LightGBM

LANFENG WANG*, YANHUI GUO*, ZELIN ZHANG, MENG'EN QIN, ZIXIN LI, XIAOLI SUN, XIAOHUI YANG†

MATCH Commun. Math. Comput. Chem.
2025

OCS-TGBM: Intelligent Analysis of Organic Chemical Synthesis Based on Topological Data Analysis and LightGBM

YANHUI GUO*, LICHAO PENG*, ZIXIN LI, MENG'EN QIN, XUE JIAO, YUN CHAI AND XIAOHUI YANG†

MATCH Commun. Math. Comput. Chem.
2024

Patents

Information Intelligent Compensation and Compression Based on Progressive Fusion of Spatial and Channel Features

LICHAO PENG, MENG'EN QIN, XIAOHUI YANG, CHEN MIAO, YANFENG SUN, HAN QIAO

CN119131792B

2025

Coal and Gangue Intelligent Recognition Based on Selective State Space Equation

XIAOHUI YANG, MENG'EN QIN, YANNI ZHANG, ZHENG GE, YUNFEI BAI

CN119048740A

2024

Intelligent Analysis Method of Stomatal Phenotype of Crop Non-destructive Leaves Based on HDIoU

LICHAO PENG, MENG'EN QIN, XIAOHUI YANG, CHEN MIAO, YANFENG SUN

CN118691972A

2024

Multi-modal Data Integration Based on Convolutional Sparse Coding and Optimal Transport

XIAOHUI YANG, JINGJING LI, YUAN FENG, LICHAO PENG, YU SONG, MENG'EN QIN

CN117763499B

2023

Software

GxEStat

MENG'EN QIN, XIAOHUI YANG, YANFENG SUN, CHEN MIAO

R

2025

- GxEStat is a user-friendly R GUI package designed for statistical analysis of genotype-by-environment interaction. Additionally, GxEStat is packaged as a standalone desktop application using Electron, and you can download the Windows release here.

Vehicle Trajectory Analysis Based on Path Planning

MENG'EN QIN

Matlab

2024

- This software integrates functionalities such as speed calculation, road travel time analysis, traffic flow estimation, congestion visualization, and so on, which are suitable for urban traffic management, intelligent transportation systems, and autonomous driving path research.

Extracurricular Activities

Stellar Corps, Young Sprout Association

Yuqing Middle School, Guizhou, China

VICE CAPTAIN AND TEACHING VOLUNTEER

Jan 2023 - Dec 2024

- I joined the Young Sprout Association of Henan University and served as vice captain of Stellar Corps (chapter for high school students). Our team brought together members from diverse disciplinary backgrounds, including medicine, chemistry, physics, education, business, and statistics. We collaborated with Shanghai Lingqing Philanthropy Development Center and co-launched a long-term volunteer project with Yuqing Middle School in Baini Town, Yuqing County, Guizhou Province. We spent most vacations (winter, summer, and national vacations) on volunteer teaching activities, student mentoring, and course design. Our team has created a series of engaging extracurricular programs that integrated academic concepts with real-world challenges.

Debate Team of School of Mathematics and Statistics, Henan University

Kaifeng, China

FOURTH DEBATER

Oct 2021 - Dec 2022

- I served as the fourth debater on the debate team of School of Mathematics and Statistics. As the closing speaker, I was responsible for delivering final rebuttals, synthesizing arguments, and constructing impactful conclusions.